

Simone Carletti

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About Me

I am a **Computer Engineering M.Sc.** student specializing in **robotics and autonomous systems**. I'm currently completing my master's thesis at EPFL, where I'm developing adaptive control for **morphing-wing drones** to achieve **agile indoor flight** using **L1** and **Reinforcement Learning**. I have hands-on experience across the robotics stack, from **perception** and **planning** to **control systems** and **AI**, working with **ROS2** and **embedded systems**. I'm proficient in **C++**, **Python**, and **TypeScript**, with research experience at EPFL and DTU. Outside the lab, I enjoy dancing salsa, hiking, taking pictures, and exploring new music.

Education

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| EPFL - LIS , <i>Master's Thesis</i> | <i>Sept 2025 – Present</i> |
| <ul style="list-style-type: none">◦ Title: L1-Adaptive augmentation for robust morphing-wing drone flight control◦ Skills: Control Systems, RL (PPO), C++, Python, Simulink, ROS2, Git, Linux, Unity | |
| DTU , <i>M.Sc. Autonomous Systems</i> | <i>Sept 2024 – Jul 2025</i> |
| <ul style="list-style-type: none">◦ Erasmus+ exchange - Converted GPA: 30/30 | |
| Politecnico di Torino , <i>M.Sc. Computer Engineering</i> | <i>Sept 2023 – Present</i> |
| <ul style="list-style-type: none">◦ Curriculum in Automation and Intelligent Cyber-Physical Systems - Current GPA: 29.48/30 | |
| Università degli Studi di Firenze , <i>B.Sc. Computer Science and Engineering</i> | <i>Sept 2020 – Sept 2023</i> |
| <ul style="list-style-type: none">◦ Graduated with 110/110 cum laude (with honors)◦ Thesis: Sensor network to monitor the greenhouses of the Botanical Garden of Florence | |

Experience

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| Robotics Engineer , <i>DIANA student team</i> | <i>Sept 2023 – Jul 2024</i>
<i>Turin, Italy</i> |
| <ul style="list-style-type: none">◦ Collaborated on a Mars rover student project, implementing perception and EKF state estimation in C++ while contributing to software development, testing, and system integration | |
| Full-Stack Developer , <i>Genomedics S.r.l</i> | <i>Sept 2020 – Present</i>
<i>Florence, Italy</i> |
| <ul style="list-style-type: none">◦ Lead developer of three key projects that handle the release, deployment, and licence management for all the company's products. Developer for the company's main project (KPlatform).◦ Skills: Angular (Typescript, SCSS, HTML), Vue.js, PHP, Laravel, SQL, Python, Git, Linux | |

Selected Projects

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| Autonomous Navigation & Task Execution - DTU RoboCup | <i>Feb 2025 – May 2025</i> |
| <ul style="list-style-type: none">◦ Developed autonomous navigation, control, and task execution algorithms with the Robobot platform for the DTU RoboCup competition | |
| Digital Control - Autonomous Robotics | <i>Oct 2024 – Dec 2024</i> |
| <ul style="list-style-type: none">◦ Designed digital controllers and implemented them on embedded microcontrollers (GitHub 🔗) | |
| Comparison and implementation (from scratch) of YOLOv1 and U-Net | <i>Oct 2024 – Dec 2024</i> |
| <ul style="list-style-type: none">◦ Implemented landmark computer vision models from scratch using PyTorch (GitHub 🔗) | |
| Localization with EKF in ROS2 | <i>Nov 2023 – Dec 2023</i> |
| <ul style="list-style-type: none">◦ Implemented a full EKF localization pipeline from scratch in ROS2 using Python | |

Awards & Activities

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| TuttoMeritoMio Excellence Scholarship | <i>2020 - 2023</i> |
| <ul style="list-style-type: none">◦ Scholarship dedicated to the top 100 students from the University of Florence | |
| ESN (Copenhagen and DTU) Volunteer | <i>Jan 2025 - Jul 2025</i> |
| <ul style="list-style-type: none">◦ Organized events, supported international students and promoted cultural exchange. | |

Skills

- **Languages:** Fluent in English, native Italian speaker, basic Spanish skills
- **Programming:** TypeScript, JavaScript, C, C++, Python, Rust, Java, Matlab, PHP, SQL, HTML, CSS/SCSS, Assembly, LaTeX
- **Tools:** PyTorch, Google Colab, OpenCV, Onshape, CMake, Docker, Node-RED, Multisim
- **Other:** Raspberry Pi, Arduino, ESP8266, NodeMCU, Soldering, Nvidia, Jetson/Orin, Guitar, Photography (digital & analog)